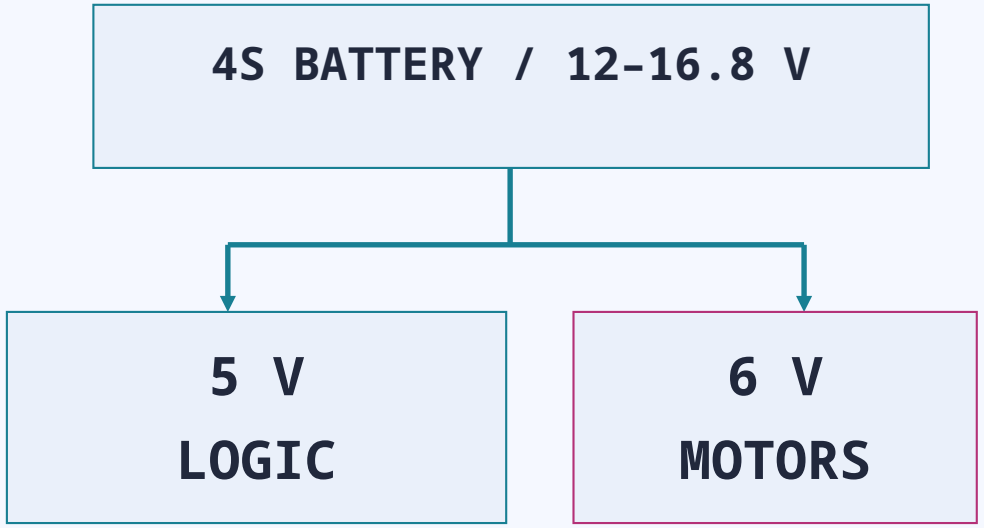
Power Architecture and ADC

Dual Buck Calibration and Analog Battery Sensing

Session 03



REGULATE / MEASURE / FILTER

Contents

01	**Power Architecture**	Separate supply rails and shared return paths
02	**Buck Calibration**	Set and verify the 5 V and 6 V outputs
03	**Analog Sensing**	Scale, convert, and filter battery voltage
04	**Implementation**	Integrate firmware, measure, and debug

Source: course PDF, pp. 1–19

One Battery, Two Different Loads

Digital logic	ESP32 and sensors need a stable supply for computation, sensing, and communication.
Motor drive	Starting, reversing, and stalling motors produce large changes in current demand.
Shared supply risk	Voltage sag and ground bounce can disturb measurements or cause the MCU to reset.

Design the power path around the load behavior.

Source: course PDF, pp. 1–2

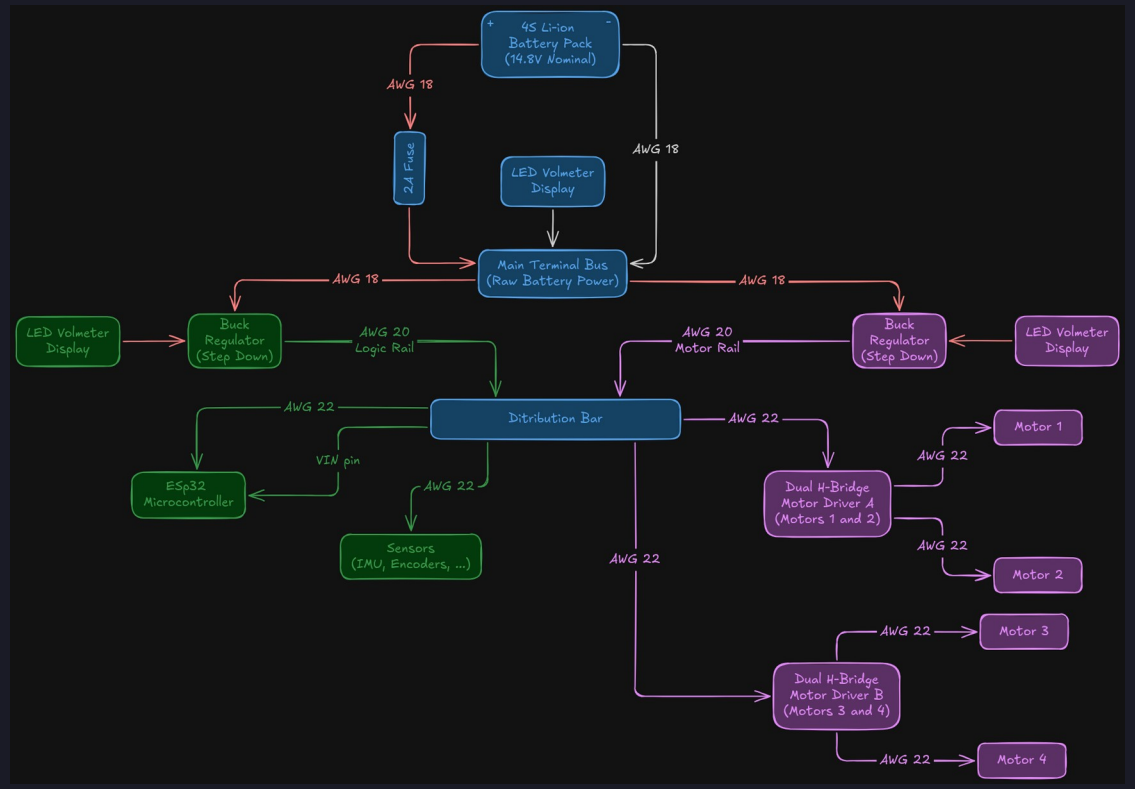
Why Motor Startup Can Reset the MCU

Current step	Starting or stalling a motor increases current through the battery and shared wiring.
Voltage drop	Battery resistance and connection impedance turn that current step into supply voltage sag.
Logic consequence	If the MCU supply drops too far, operation becomes unreliable and a reset may occur.

Two regulators reduce coupling; they do not remove shared-path impedance.

Source: course PDF, pp. 1–2

Power Architecture: Original Overview



FOLLOW THE POWER

Battery → fuse → bus

Two buck branches

Logic and motor loads

The distribution assembly carries separate 5 V and 6 V positive rails.

Source: course PDF, pp. 1

Trace Each Power Rail

Battery bus	The fused battery positive feeds both buck inputs; raw battery voltage is 12–16.8 V.
5 V logic rail	Buck 1 supplies ESP32 VIN and suitable 5 V loads. It does not connect to the 3V3 pin.
6 V motor rail	Buck 2 supplies driver VM in this course example; check the actual motor voltage rating.

Keep the two positive outputs separate; connect their returns to the ground bus.

Source: course PDF, pp. 1–2

The 4S Battery Operating Range

FULLY CHARGED	NOMINAL	LAB THRESHOLD
16.8 V	**14.8 V**	**12.0 V**
4 × 4.2 V Maximum pack voltage.	4 × 3.7 V Nominal rating, not a constant voltage.	4 × 3.0 V Pack-level low-voltage warning threshold.

Pack voltage alone does not reveal individual cell imbalance.

Source: course PDF, pp. 2

Four Cells in Series: Voltage Adds

$$V_{\text{pack}} = V_1 + V_2 + V_3 + V_4$$

$$V_{\text{full}} = 4 \times 4.2\,\text{V} = 16.8\,\text{V}$$

$$V_{\text{nom}} = 4 \times 3.7\,\text{V} = 14.8\,\text{V}$$

4S1P means four cells in series and one parallel string.

Source: course PDF, pp. 2

Buck Regulator Input Headroom

$$V_{\text{in, min}} \geq V_{\text{out}} + V_{\text{dropout}}$$

$$V_{\text{in, min}} \geq 6.0\,\text{V} + 1.5\,\text{V} = 7.5\,\text{V}$$

$$12.0\,\text{V} - 7.5\,\text{V} = 4.5\,\text{V}$$

This is the course headroom example; verify the actual module under load.

Source: course PDF, pp. 2

Protection and Return Paths

Fuse near battery	The reference uses a 2 A inline fuse. Confirm its rating against wiring and load demand.
Star ground	Return high motor currents directly to the main terminal bus using suitable heavy wire.
Logic wiring	Keep motor current out of breadboard tracks and thin logic jumpers; use local decoupling.

A fuse limits fault energy; its clearing time depends on fault current.

Source: course PDF, pp. 2–3

Fuse and Star Bus: Physical Placement

Battery positive	Place the reference 2 A fuse near the battery before the main positive distribution terminal.
Battery negative	Use the main negative terminal as the common reference for the return branches.
Motor return	Run suitable heavy wire from the drivers to the bus; do not pass it through logic wiring.

Reference wiring uses AWG 18 for the main feed and AWG 18–20 for returns.

Source: course PDF, pp. 2–3

Before Applying Power

Disconnect loads	Keep ESP32 VIN and motor-driver VM disconnected until both rails are adjusted.
Inspect + measure	With power OFF, verify polarity, fuse, ground continuity, and no persistent rail short.
Identify outputs	Label Buck 1 = 5 V and Buck 2 = 6 V. Use adjustable modules for these setpoints.

Use resistance and continuity modes only on an unpowered circuit.

Source: course PDF, pp. 7

Measure the Battery Before Adjustment

Meter mode	Select DC voltage; use a range above the maximum pack voltage of 16.8 V.
Probe placement	Black probe → main negative terminal. Red probe → fused battery-positive bus.
Cross-check	Read the meter and the battery display. The course comparison target is ±0.10 V.

Confirm a positive reading and correct polarity before adjusting either buck.

Source: course PDF, pp. 7

Calibrate Buck 1: The 5 V Logic Rail

Disconnect loads	Leave ESP32 VIN and the logic loads disconnected during initial adjustment.
Measure output	Probe OUT+ relative to OUT– on Buck 1. Keep the probes clear of neighboring terminals.
Adjust + compare	Turn the trimmer while watching the meter. Set 5.00 V and compare the 5 V display.

Use the meter as the reference; trimmer direction depends on the module.

Source: course PDF, pp. 7

Calibrate Buck 2: The 6 V Motor Rail

Disconnect drivers	Keep motor-driver VM disconnected while setting the output voltage.
Measure output	Probe Buck 2 OUT+ relative to OUT–. Adjust carefully until the meter reads 6.00 V.
Label + verify	Compare the motor-rail display and label the output before reconnecting the driver.

After reconnecting, check the output during startup and normal motor loading.

Source: course PDF, pp. 7

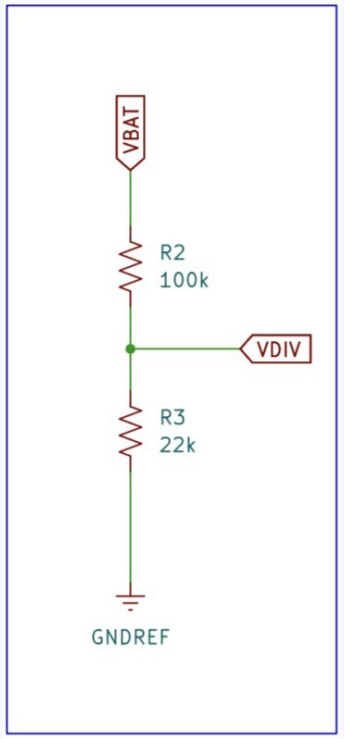
Verify Before Connecting the Loads

Power OFF	Verify that 5 V and 6 V positives are not accidentally tied together; grounds are shared.
Reconnect	Connect each load to its labeled rail. Recheck polarity before applying power.
Power ON	Measure both rails under load and during motor startup; investigate sag or overheating.

A resistance reading alone does not prove electrical isolation.

Source: course PDF, pp. 7–9

Resistive Divider: Circuit and Equation



$$V_{\text{pin}} = V_{\text{bat}} \frac{R_{\text{bottom}}}{R_{\text{top}} + R_{\text{bottom}}}$$

$$R_{\text{top}} = 100\,\text{k}\Omega, \quad R_{\text{bottom}} = 22\,\text{k}\Omega$$

VDIV is the midpoint connected to GPIO 32.

R2 is above VDIV; R3 is below it.

Use top/bottom names in the calculations; the schematic labels these R2 and R3.

Source: course PDF, pp. 3–5, 8

Calculate the Divider Ratio

$$K_{\text{div}} = \frac{R_{\text{bottom}}}{R_{\text{top}} + R_{\text{bottom}}}$$

$$K_{\text{div}} = \frac{22\,\text{k}\Omega}{100\,\text{k}\Omega + 22\,\text{k}\Omega}$$

$$K_{\text{div}} = \frac{22}{122} \approx 0.1803279$$

The ADC receives about 18.03% of the battery voltage.

Source: course PDF, pp. 4–5

Check the Divider at Both Limits

$$V_{\text{pin, max}} = 16.8\,\text{V} \times \frac{22}{122} \approx 3.0295\,\text{V}$$

$$V_{\text{pin, min}} = 12.0\,\text{V} \times \frac{22}{122} \approx 2.1639\,\text{V}$$

Measure the assembled midpoint before connecting it to the ADC.

Source: course PDF, pp. 5

Divider Current and ADC Headroom

$$I_{\mathrm{div}} = \frac{V_{\mathrm{bat}}}{R_{\mathrm{top}} + R_{\mathrm{bottom}}} = \frac{16.8\,\mathrm{V}}{122\,\mathrm{k\Omega}} \approx 137.7\,\mu\mathrm{A}$$

$$3.10\,\mathrm{V} - 3.0295\,\mathrm{V} = 70.5\,\mathrm{mV}$$

$$V_{\mathrm{pin,\,max,\,1\%}} = 16.8\,\mathrm{V}\frac{22.22}{99 + 22.22} \approx 3.0795\,\mathrm{V}$$

1% tolerance example: top −1%, bottom +1%. Check the actual ADC input range.

Source: course PDF, pp. 5

Choose an ADC1 Input

GPIO 32	ADC1 channel 4; a general-purpose I/O. Configure it as an input for battery sensing.
GPIO 34	ADC1 channel 6; input-only. No internal pull-up or pull-down resistor.
Wi-Fi coexistence	Prefer ADC1 for this classic ESP32 design. ADC2 is shared with the Wi-Fi subsystem.

Pin mappings differ across ESP32 variants; check the actual board.

Source: course PDF, pp. 5

ADC Resolution and Input Range

RESOLUTION	ATTENUATION	PRACTICAL CHECK
12 bits	**ADC_11db**	**3.030 V**
2^{12} = 4096 levels Raw codes: 0–4095.	Classic ESP32 documented range: about 0.15–3.10 V. Check device variant.	The nominal full-pack divider voltage is close to the top of this range.

Resolution, measurable range, and accuracy are different properties.

Source: course PDF, pp. 5-6

Quantization: The Ideal 3.3 V Model

$$2^{12} = 4096 \text{ levels}, \qquad N_{\mathrm{ADC}} \in \{0, \ldots, 4095\}$$

$$q_{\mathrm{pin}} = \frac{3.3\,\mathrm{V}}{4096} \approx 0.80566\,\mathrm{mV/count}$$

$$q_{\mathrm{battery}} = q_{\mathrm{pin}} \frac{122}{22} \approx 4.4678\,\mathrm{mV/count}$$

An ideal 12-bit quantizer has 4096 bins; this is not an ESP32 calibration model.

Source: course PDF, pp. 5–6

Undo the Divider: Reconstruct Voltage

$$K_{\text{inv}} = \frac{R_{\text{top}} + R_{\text{bottom}}}{R_{\text{bottom}}} = \frac{122}{22} \approx 5.54545$$

$$V_{\text{pin}} = 2669\,\text{mV} = 2.669\,\text{V}$$

$$V_{\text{bat}} = V_{\text{pin}} K_{\text{inv}} = 2.669\,\text{V} \times \frac{122}{22} \approx 14.80\,\text{V}$$

Firmware reads calibrated millivolts, then applies the external divider ratio.

Source: course PDF, pp. 6, 11–12

Raw Counts: The Course Endpoint Approximation

$$V_{\text{pin}} \approx \frac{N_{\text{ADC}}}{4095} \times 3.3\,\text{V}$$

$$V_{\text{bat}} \approx \frac{N_{\text{ADC}}}{4095} \times 3.3\,\text{V} \times \frac{122}{22}$$

$$V_{\text{bat}} \approx N_{\text{ADC}} \times 0.0044689\,\text{V}$$

Dividing by 4095 maps both endpoints; it differs from the 4096-bin quantization step.

Source: course PDF, pp. 6

Why Raw Voltage Readings Move

Switching ripple	Buck converters inject switching components onto the supply and ground network.
Motor transients	Commutation and changing load current cause spikes, voltage sag, and ground bounce.
Measurement chain	ADC noise, divider tolerances, and wiring add variation to the sampled voltage.

Improve wiring first; filtering cannot repair a bad electrical connection.

Source: course PDF, pp. 6

Exponential Moving Average (EMA)

$$y[k] = \alpha x[k] + (1 - \alpha) y[k-1]$$

$$\alpha = 0.15 \quad \Rightarrow \quad y[k] = 0.15 x[k] + 0.85 y[k-1]$$

x[k]: new sample; y[k – 1]: previous result; y[k]: updated result.

Source: course PDF, pp. 6–7, 12

Work Through One Filter Update

$$x[k] = 14.2\,\mathrm{V}, \qquad y[k-1] = 14.8\,\mathrm{V}$$

$$y[k] = 0.15(14.2\,\mathrm{V}) + 0.85(14.8\,\mathrm{V})$$

$$y[k] = 2.13\,\mathrm{V} + 12.58\,\mathrm{V} = 14.71\,\mathrm{V}$$

Worked example: one low sample moves the filtered value by only 0.09 V.

Source: course PDF, pp. 6–7, 12

Choose the Smoothing Coefficient

α = 1.00

Fast / noisy

Output equals the newest sample.
No smoothing.

α = 0.15

Lab starting point

Balances smoothing and responsiveness at the 10 Hz sample rate.

α = 0.01

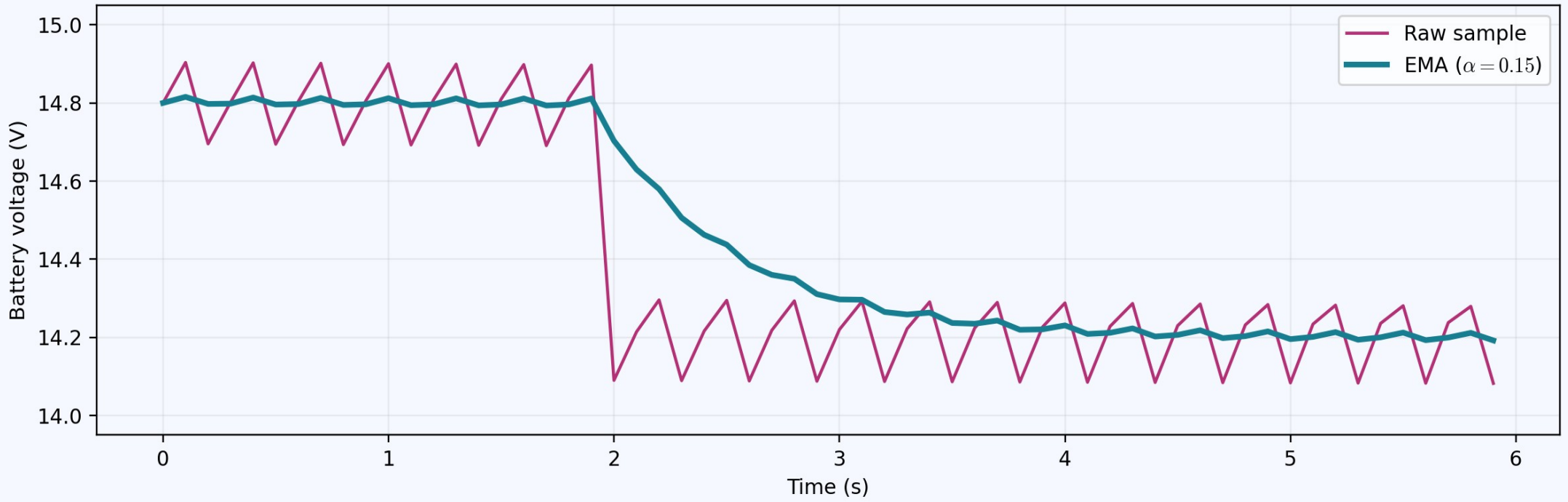
Slow / smooth

Strong smoothing with substantial lag after voltage changes.

The response depends on both α and the sampling interval.

Source: course PDF, pp. 6–7

See the Effect of the Filter



Illustrative 10 Hz samples: smoothing reduces variation but adds response lag.

Source: course PDF, pp. 6–7

Battery Percentage and Low Voltage

$$P = \mathrm{clamp}\left(100\frac{V_{\mathrm{filtered}} - 12.0\,\mathrm{V}}{16.8\,\mathrm{V} - 12.0\,\mathrm{V}},\ 0,\ 100\right)\%$$

$$V_{\mathrm{filtered}} < 12.0\,\mathrm{V} \quad \Rightarrow \quad \text{low battery warning}$$

This is a voltage indicator, not true state of charge; the code reports a warning.

Source: course PDF, pp. 12–15

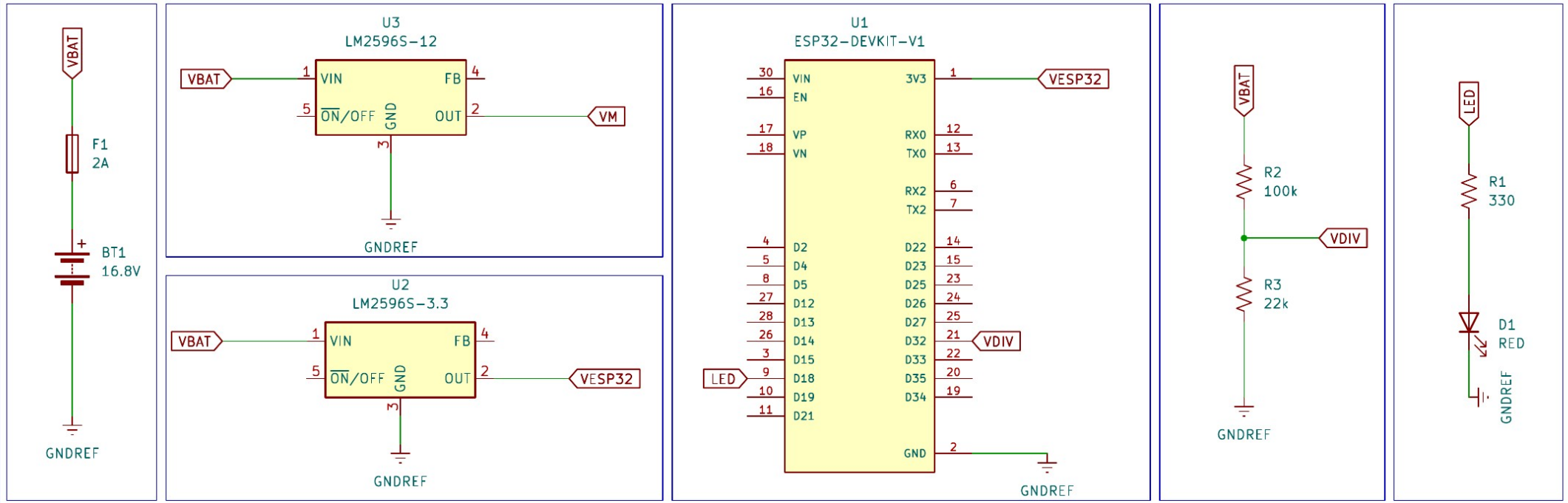
Connect the Measurement Chain

Power	Buck 1 → ESP32 VIN / suitable 5 V loads. Buck 2 → motor-driver VM terminals.
Sense	Battery → 100 kΩ → GPIO 32 midpoint. Midpoint → 22 kΩ → common ground.
Outputs + serial	GPIO 18: PWM; GPIO 4: heartbeat LED. Use current limiting for LEDs; serial: 115200 baud.

Keep the 5 V and 6 V rails separate and verify the board power input.

Source: course PDF, pp. 8–10, 15–17

Power System: Full Reference Schematic

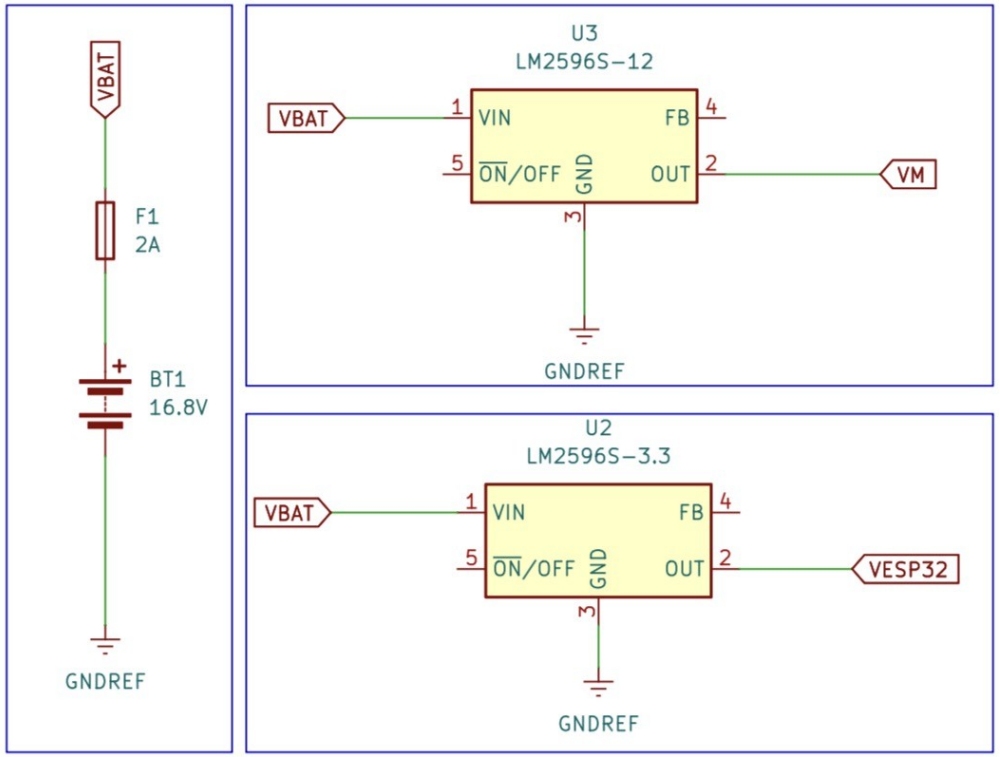


Original 3.3 V / 12 V drawing; the lab uses 5 V → VIN and a separate motor rail.

Read VBAT, VM, VDIV, LED, and GNDREF as connected nets across the sheet.

Source: course PDF, pp. 8–9

Schematic: Battery, Fuse, and Buck Supplies



VBAT + F1

Battery feed through the 2 A reference fuse.

U2 / U3

Both converter branches share GNDREF.

READ THE LABELS

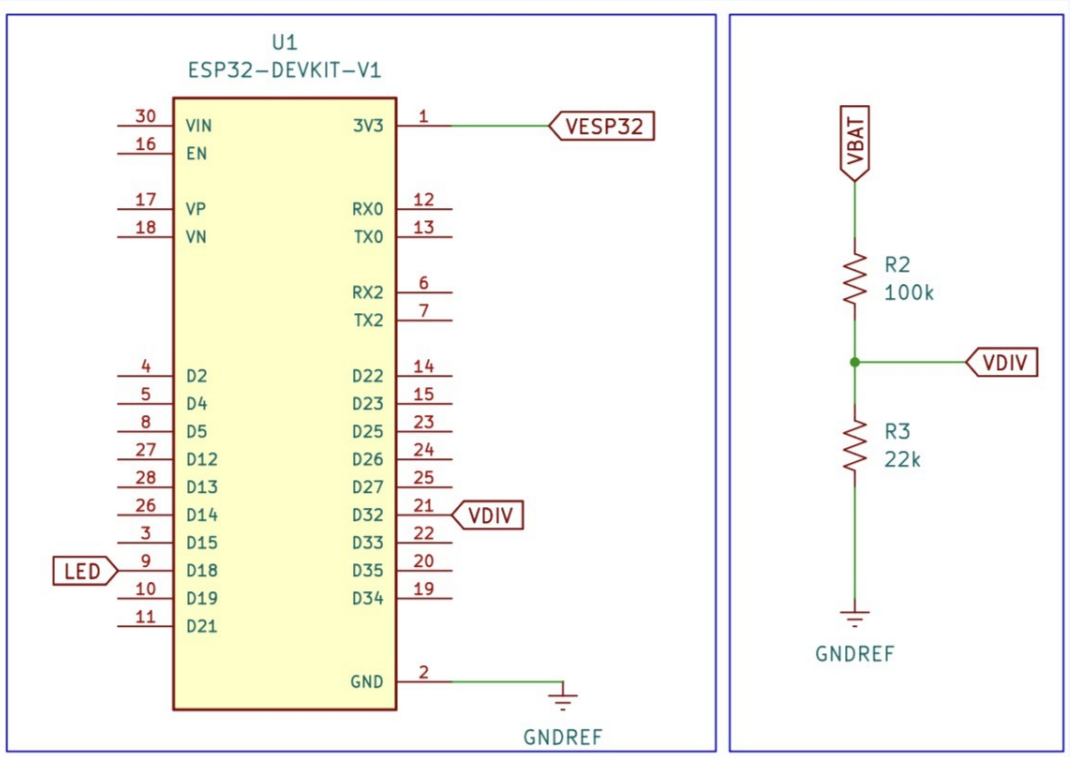
Drawing: 3.3 V / 12 V.

Lab procedure: 5 V / 6 V.

Use adjustable modules for the lab setpoints; the original labels are shown unchanged.

Source: course PDF, pp. 9

Schematic: ESP32 and Battery Sensing



R2 / R3

100 kΩ above VDIV;
22 kΩ below VDIV.

VDIV → D32

The divider midpoint feeds GPIO 32 / ADC1.

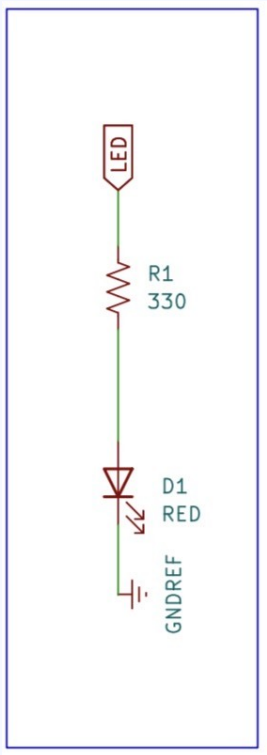
LAB POWER INPUT

Connect the 5 V rail to VIN, not to 3V3.

The shown 3V3 connection belongs to the drawing's 3.3 V supply version.

Source: course PDF, pp. 8

Schematic: PWM LED and Current Limiting



LED NET

The core drawing maps this net to GPIO 18.

R1 = 330 Ω

Series resistor limits current through D1.

D1 → GNDREF

Confirm LED polarity and the return path.

This supplied image shows the PWM LED circuit; the heartbeat uses GPIO 4.

Source: course PDF, pp. 9

Extend the Three-Rate Scheduler

MOTION

100 Hz

10 ms interval
Update PWM duty
and timing metrics.

TELEMETRY

10 Hz

100 ms interval
Read ADC, filter,
and report voltage.

HEARTBEAT

1 Hz task

1 second interval
Toggle the LED
on GPIO 4.

Hardware PWM continues at 20 kHz while the loop schedules tasks.

Source: course PDF, pp. 10, 13–17

Configure the Battery Measurement

```
const uint8_t PIN_BATTERY_ADC = 32;
const float R_TOP_OHMS = 100000.0f;
const float R_BOTTOM_OHMS = 22000.0f;
const float DIVIDER_SCALE =
  (R_TOP_OHMS + R_BOTTOM_OHMS) / R_BOTTOM_OHMS;
const float FILTER_ALPHA = 0.15f;
const float V_BAT_EMPTY = 12.0f;
```

Keep the pin, divider scale, filter coefficient, and warning threshold together.

Source: course PDF, pp. 10

Read Calibrated Battery Voltage

```
float readBatteryVoltage() {
  const uint32_t pin_voltage_millivolts =
    analogReadMilliVolts(PIN_BATTERY_ADC);
  const float pin_voltage =
    pin_voltage_millivolts / 1000.0f;
  return pin_voltage * DIVIDER_SCALE;
}
```

Convert calibrated millivolts to volts, then undo the resistor divider.

Source: course PDF, pp. 11–12

Initialize the ADC and Prime the Filter

```
void initBatterySensing() {
  pinMode(PIN_BATTERY_ADC, INPUT);
  analogReadResolution(12);
  analogSetPinAttenuation(PIN_BATTERY_ADC, ADC_11db);

  raw_battery_voltage = readBatteryVoltage();
  filtered_battery_voltage = raw_battery_voltage;
}
```

Call once in setup(); initialize the filtered state from a real sample.

Source: course PDF, pp. 12

Filter and Check the Battery

```
void updateBatterySensing() {
  raw_battery_voltage = readBatteryVoltage();
  filtered_battery_voltage =
    FILTER_ALPHA * raw_battery_voltage +
    (1.0f - FILTER_ALPHA) * filtered_battery_voltage;
  low_battery_flag =
    filtered_battery_voltage < V_BAT_EMPTY;
}
```

Update the EMA and evaluate the warning against the filtered voltage.

Source: course PDF, pp. 12–13

Report the Two Voltage Traces

```
void taskTelemetry() {
  updateBatterySensing();
  Serial.printf(
    "V_bat_Raw: %.2f,V_bat_Filt: %.2f\n",
    raw_battery_voltage, filtered_battery_voltage);
}
```

Focused plot output: compare the raw and filtered voltage at 10 Hz.

Source: course PDF, pp. 14

Report the Low-Battery Condition

```
if (low_battery_flag) {
  Serial.println(
    "[ALARM] LOW BATTERY VOLTAGE BELOW 12.0 V"
  );
}
```

Add inside the telemetry task for Serial Monitor use; omit text alarms while plotting.

Source: course PDF, pp. 14–15

Schedule Battery Sensing at 10 Hz

```
if ((uint32_t)(current_time_ms - last_telemetry_time_ms)
    >= INTERVAL_TELEMETRY_MS) {

  last_telemetry_time_ms += INTERVAL_TELEMETRY_MS;
  taskTelemetry();
}
```

Inside loop(): a 100 ms interval, independent of motion and heartbeat timing.

Source: course PDF, pp. 17

Bench Test: Meter, Monitor, Plotter

Measure	Probe the battery terminals and GPIO 32. Compare with the expected voltages on the next slide.
Observe	At 115200 baud, expect 10 samples per second. Plot raw and filtered voltage together.
Compare	Aim for agreement within ±0.10 V at the pack after checking divider values and calibration.

Treat ±0.10 V as the lab target, not a guaranteed ADC specification.

Source: course PDF, pp. 17–18

Expected Bench Measurements

FULL PACK	NOMINAL PACK	WARNING LEVEL
16.80 V	**14.80 V**	**12.00 V**
Divider midpoint ≈ 3.030 V	Divider midpoint ≈ 2.669 V	Divider midpoint ≈ 2.164 V

The filtered value should settle near the meter reading.

Source: course PDF, pp. 5, 18

Debugging: Wrong Voltage

Reads zero	Check the divider midpoint connection, GPIO number, ground path, and resistor joints.
Wrong scale	Verify 100 kΩ / 22 kΩ orientation, measured resistor values, and the 5.54545 scale factor.
Clips near full	Check attenuation and the chip input range; compare the physical midpoint voltage first.

Measure the signal chain in order: pack → divider → ADC → software.

Source: course PDF, pp. 18

Debugging: Noise, Lag, and Alarms

Noisy voltage	Inspect motor return routing and connections. Reduce α if smoothing is insufficient.
Slow tracking	Increase α cautiously from 0.15 toward 0.3; confirm the sampling interval is still 100 ms.
Alarm on startup	Compare raw and filtered sag with a meter or scope; check battery and wiring impedance.

Do not hide persistent undervoltage by adding more filtering.

Source: course PDF, pp. 18–19

Lab Completion Checklist

Power verified	5 V logic and 6 V motor rails measured with correct polarity and separate outputs.
Sensing verified	Divider midpoint and reconstructed pack voltage agree with physical measurements.
Software verified	Stable 10 Hz telemetry, primed EMA, and the low-voltage condition understood.

Explain each measurement before moving to motor control.

Source: course PDF, pp. 7, 17–19

References / 1 of 2

01 The Art of Electronics: Student Manual
Thomas C. Hayes and Paul Horowitz

02 Making Embedded Systems: Design Patterns for Great Software
Elecia White

03 Feedback Systems: An Introduction for Scientists and Engineers
Karl Johan Åström and Richard M. Murray

04 ADC Peripheral API Documentation — Espressif Systems

Source: course PDF, pp. 19

References / 2 of 2

05 GPIO Peripheral API Documentation — Espressif Systems

06 Espressif Centralized Documentation Platform (CDP)

07 Arduino Core for ESP32 Documentation — Espressif Systems

08 Arduino Language Reference — Arduino

Source: course PDF, pp. 19

Thank you!

Questions & discussion

